

STABLE HOMOTOPY CATEGORIES AND STABLE HOMOTOPY GROUPS

July 17, 2026

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The purpose of these notes is to concisely define stable homotopy categories and to state their relationship to stable homotopy groups.

1. DEFINITIONS

All spaces are assumed to be compactly generated.

X, Y : (Compactly generated) spaces. Unless otherwise specified, they will be based spaces. By default, $*$ denotes the base point of a based space.

Definition 1.0.1. Let X and B be topological spaces (Definition A.0.1) and let $p : X \rightarrow B$ be a continuous map (Definition A.0.2).

The map p is called a *fibration* or a *Serre fibration* if it satisfies the *Homotopy Lifting Property (HLP)*: for every space Y , every map $f : Y \rightarrow X$, every homotopy $H : Y \times I \rightarrow B$ with $H(y, 0) = p(f(y))$ for all $y \in Y$, and whenever a lift $\tilde{f}_0 : Y \rightarrow X$ of $H(\cdot, 0)$ is given, there exists a homotopy $\tilde{H} : Y \times I \rightarrow X$ such that

$$\tilde{H}(y, 0) = \tilde{f}_0(y) \quad \text{for all } y \in Y, p \circ \tilde{H} = H \quad \text{on } Y \times I.$$

Equivalently, for all Y , the map

$$\text{Map}(Y, X) \rightarrow \text{Map}(Y, B), \quad f \mapsto p \circ f$$

has the right lifting property with respect to the inclusion $\text{Map}(Y, *) \hookrightarrow \text{Map}(Y \times I, B)$ encoded by the HLP.

Definition 1.0.2. Let X be a topological space and let $A \subseteq X$ be a subspace. The inclusion map $i : A \hookrightarrow X$ is called a **cofibration** if it satisfies the **Homotopy Extension Property (HEP)**: for every space Y , every map $f : A \rightarrow Y$ and every homotopy $H : X \times I \rightarrow Y$ with $H(a, t) = f(a)$ for all $a \in A$ and all $t \in I$, there exists a homotopy $\tilde{H} : X \times I \rightarrow Y$ extending H such that $\tilde{H}|_{A \times I} = H|_{A \times I}$ and $\tilde{H}(x, 0) = f(x)$ for all $x \in A$. Equivalently, the inclusion $i : A \hookrightarrow X$ has the left lifting property with respect to every map that is a fibration.

Definition 1.0.3. A pointed topological space (Definition A.0.6) X is said to be **nondegenerately based** or **well pointed** if the inclusion $* \hookrightarrow X$ is a cofibration (Definition 1.0.2) in the unbased sense, i.e. the inclusion satisfies the homotopy extension property.

Definition 1.0.4 (Discrete topological space). Let X be a set. The **discrete topology on X** is the topology $\mathcal{T}_{\text{disc}}$ defined by

$$\mathcal{T}_{\text{disc}} := \{U : U \subseteq X\},$$

i.e., every subset of X is open. The pair $(X, \mathcal{T}_{\text{disc}})$ is called a **discrete topological space** or even a **discrete set**.

Definition 1.0.5 (Indiscrete (or trivial) topological space). Let X be a set. The **indiscrete topology** (also called the **trivial topology**) on X is the topology $\mathcal{T}_{\text{indisc}}$, also denoted by notations such as $\mathcal{T}_{\text{triv}}$, defined by

$$\mathcal{T}_{\text{indisc}} := \{\emptyset, X\},$$

i.e., only the empty set and the entire set are open. The pair $(X, \mathcal{T}_{\text{indisc}})$ is called an **indiscrete topological space**.

Proposition 1.0.6 ([May99, Proposition Page 41]). For compactly generated (Definition A.0.10) topological spaces (Definition A.0.1) X, Y, Z , the canonical bijection

$$Z^{(X \times Y)} \cong (Z^Y)^X$$

of function spaces (with the compact-open topology) (Definition A.0.27) is a homeomorphism (Definition A.0.8).

Proposition 1.0.7. For compactly generated (Definition A.0.10) pointed topological spaces (Definition A.0.6) X, Y, Z , the canonical bijection

$$\text{Map}_*(X \wedge Y, Z) \rightarrow \text{Map}_*(X, \text{Map}_*(Y, Z))$$

(Definition A.0.26) of function spaces (with the compact-open topology) (Definition A.0.27) is a homeomorphism (Definition A.0.8).

$F(X, Y)$: The subspace of Y^X consisting of the based maps, with the constant based map as basepoint. We have a natural homeomorphism

$$F(X \wedge Y, Z) \cong F(X, F(Y, Z))$$

of based spaces for any based spaces X, Y , and Z .

Definition 1.0.8 (Homotopy groups of a pointed topological space). For any pointed topological space (Definition A.0.6) (X, x_0) and integer $n \geq 0$, the **n -th homotopy set of X at x_0** , denoted $\pi_n(X, x_0)$ or simply $\pi_n(X)$ when the base point x_0 is understood or a choice

of base point does not really matter, is defined as the set of all homotopy classes (rel. ∂I^n) (Definition A.0.7) of paired maps

$$f : (I^n, \partial I^n) \rightarrow (X, \{x_0\}),$$

where $I^n = [0, 1]^n$ and ∂I^0 is taken to be the singleton I^0 instead of the literal boundary thereof. Equivalently, one can also define it as the set of all homotopy classes of pointed maps (Definition A.0.6)

$$f : (S^n, s_0) \rightarrow (X, x_0),$$

where S^n is the n -sphere and $s_0 \in S^n$ is a choice of base point. In other words, $\pi_n(X, x_0)$ is by definition $[(S^n, s_0), (X, x_0)]_*$ (Definition A.0.7).

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of pointed maps, and for $n \geq 2$, it is abelian. As such, for $n \geq 1$, it is appropriate to refer to $\pi_n(X, x_0)$ as the *n th homotopy group of (X, x_0)* .

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

For every $n \geq 0$, the assignment $(X, x_0) \mapsto \pi_n(X, x_0)$ is functorial — in fact

1. For $n = 0$, it is a functor $\text{Top}_* \rightarrow \mathbf{Sets}$,
2. For $n = 1$, it is a functor $\text{Top}_* \rightarrow \mathbf{Grp}$,
3. For $n \geq 2$, it is a functor $\text{Top}_* \rightarrow \mathbf{Ab}()$.

Given a map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces, the induced map $\pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$ is often denoted by f_* .

The homotopy set $\pi_n(X, x_0)$ is naturally isomorphic to the relative homotopy set $\pi_n(X, \{x_0\}, x_0)$ (Definition A.0.21).

We further note that the set $\pi_n(X, x_0) = [(S^n, s_0), (X, x_0)]_*$ is identifiable with $\pi_0(\text{Map}_*((S^n, s_0), (X, x_0)))$ (Proposition 1.0.10).

Lemma 1.0.9. Let X be a topological space. Let $x_0 \in X$. The set $\pi_0(X, x_0)$ (Definition A.0.20) bijectively corresponds to the set of path connected components of X .

Proposition 1.0.10. Let (X, x_0) and (Y, y_0) be pointed spaces (Definition A.0.6). There is a canonical bijection

$$[X, Y]_* \cong \pi_0(\text{Map}_*(X, Y), f_0)$$

(Definition A.0.20) (Definition A.0.27) where $f_0 : X \rightarrow Y$ is the constant map to y_0 , and $\text{Map}_*(X, Y)$ is equipped with the compact open topology (Definition A.0.27).

Definition 1.0.11. A topological space (Definition A.0.1) X is said to be *n -connected* if $\pi_q(X, x) = 0$ (Definition 1.1.10) for $0 \leq q \leq n$ for all $x \in X$.

Definition 1.0.12 (Unreduced suspension). Let X be a topological space (Definition A.0.1). The *unreduced suspension of X* , denoted by SX , is the quotient space of the product $X \times [0, 1]$ obtained by collapsing $X \times \{0\}$ to a single point and $X \times \{1\}$ to another (distinct) single point.

$$SX := (X \times [0, 1]) / \sim, \quad \text{where } (x, 0) \sim (x', 0) \text{ and } (x, 1) \sim (x', 1) \text{ for all } x, x' \in X.$$

Geometrically, SX consists of two cones on X glued along their common base $X \times \{1/2\}$.
(♠ TODO: cone)

1.1. Stable homotopy groups and the stable homotopy theory.

Definition 1.1.1. Let (X, x_0) be a pointed topological space (Definition A.0.6). The *n th suspension homomorphism of X* is the function $\pi_n(X, x_0) \rightarrow \pi_{n+1}(\Sigma X, [x_0, 0])$ (Definition A.0.20), which is a group homomorphism whenever $n \geq 1$, defined by sending $[f] \in \pi_n(X, x_0)$ the homotopy class (Definition A.0.7) of

$$\Sigma f = f \wedge \text{id} : S^{n+1} \cong S^n \wedge S^1 \rightarrow X \wedge S^1.$$

(♠ TODO: continue going through here)

Theorem 1.1.2 (Freudenthal suspension, see e.g. [May99, Chapter 11]). If X is nondegenerately based (Definition 1.0.3) and $(n-1)$ -connected (Definition 1.0.11) where $n \geq 1$, then the suspension homomorphism (Definition 1.1.1) $\Sigma : \pi_q(X) \rightarrow \pi_{q+1}(\Sigma X)$ is a bijection if $q < 2n-1$ and a surjection if $q = 2n-1$.

Definition 1.1.3. A *prespectrum* T is a sequence of pointed topological space (Definition A.0.6) T_n and pointed maps (Definition A.0.6) $\sigma = \sigma_n^T : \Sigma T_n \rightarrow T_{n+1}$ (Definition A.0.28). The space T_n may be referred to as the *degree n space of the prespectrum* and the maps $\sigma : \Sigma T_n \rightarrow T_{n+1}$ may be referred to as the *structure maps*.

A *morphism $f : T \rightarrow T'$ of prespectra* is a sequence of maps $f_n : T_n \rightarrow T'_n$ such that $\sigma'_n \circ \Sigma f_n = f_{n+1} \circ \sigma_n$ for all n .

Definition 1.1.4. A *spectrum* E is a prespectrum (Definition 1.1.3) such that the adjoints $\tilde{\sigma} : E_n \rightarrow \Omega E_{n+1}$ to the structure maps $\sigma : \Sigma T_n \rightarrow T_{n+1}$ are homeomorphisms (Definition A.0.8).

A *morphism $f : E \rightarrow E'$ of spectra* is a morphism of between spectra as prespectra.

Definition 1.1.5. The *suspension spectrum of a pointed topological space X* is the spectrum (Definition 1.1.4) $\Sigma^\infty X = \Sigma^\infty(X)$ whose degree n space is $\Sigma^n X$ (Definition A.0.28).

Definition 1.1.6. Let X be a *pointed topological space*. The *q th stable homotopy group of X* is the group $\pi_q^s(X) = \pi_q(\Sigma^\infty X)$ defined as

$$\pi_q^s(X) = \pi_q(\Sigma^\infty X) := \text{colim}_n \pi_{q+n}(\Sigma^n X)$$

(Definition A.0.20) where the colimit is of the filtered direct system of the maps

$$\begin{aligned} \pi_{q+n}(\Sigma^n X) &\rightarrow \pi_{q+n+1}(\Sigma^{n+1} X) \\ [f] &\mapsto [\sigma_n \circ \Sigma f] \end{aligned}$$

where $\Sigma f : \Sigma S^{q+n} \rightarrow \Sigma(\Sigma^n X)$ is the suspension (Definition A.0.28) of the map $f : S^{q+n} \rightarrow \Sigma^n X$ and $\sigma_n : \Sigma(\Sigma^n X) \xrightarrow{\cong} \Sigma^{n+1} X$ is the canonical structure map of the suspension spectrum (Definition 1.1.5).

By the Fruedenthal Suspension theorem (Theorem 1.1.2), $\Sigma^n : \pi_q(X) \rightarrow \pi_{q+n}(\Sigma^n X)$ is an isomorphism whenever $q < n - 1$, so

$$\pi_q(\Sigma^\infty X) \cong \pi_{q+n}(\Sigma^n X)$$

for any $n > q + 1$. In particular, $\pi_q(\Sigma^\infty X)$ is abelian.

See stable homotopy theory (Definition 1.1.9), which is the reduced homology theory given (Corollary 1.1.8) by the functors $\{X \mapsto \pi_q(\Sigma^\infty X)\}_q$.

The stable homotopy groups of X coincide with the homotopy groups (Definition 1.1.10) of the suspension spectrum $\Sigma^\infty X$ (Definition 1.1.5).

Theorem 1.1.7. [May99, Theorem Chapter 22 Page 176] Let $\{T_n\}$ be a prespectrum (Definition 1.1.3) such that T_n is $(n - 1)$ -connected (Definition 1.0.11) and of the homotopy type of a CW complex for each n . Define

$$\tilde{E}_q(X) = \operatorname{colim}_n \pi_{q+n}(X \wedge T_n)$$

where the colimit is taken over the maps

$$\pi_{q+n}(X \wedge T_n) \xrightarrow{\Sigma} \pi_{q+n+1}(\Sigma(X \wedge T_n)) \cong \pi_{q+n+1}(X \wedge \Sigma T_n) \xrightarrow{\operatorname{id} \wedge \sigma} \pi_{q+n+1}(X \wedge T_{n+1}).$$

(♠ TODO: reduced homology theory) Then the functors $\tilde{E}_q$ define a reduced homology theory on based CW complexes.

Corollary 1.1.8. The stable homotopy groups (Definition 1.1.6) $\pi_q(\Sigma^\infty X)$ give a reduced homology theory. (♠ TODO: reduced homology theory)

Definition 1.1.9. (♠ TODO: reduced homology theory) The reduced homology theory given by (Corollary 1.1.8) the functors $\{X \mapsto \pi_q(\Sigma^\infty X)\}_q$ (Definition 1.1.6) is called *stable homotopy theory*.

More generally, we can define the stable homotopy group of a spectrum.

Definition 1.1.10 ([nLa25a, Definition 2.1]). Let E be a spectrum (Definition 1.1.4). For $q \in \mathbb{Z}$, the *q th (stable) homotopy group of E* is the colimit

$$\pi_q(E) := \operatorname{colim}_n \pi_{q+n}(E_n)$$

where the colimit is taken over the maps

$$\pi_{q+n}(E_n) \xrightarrow{\Sigma} \pi_{q+n+1}(\Sigma E_n) \xrightarrow{\pi_{q+n+1}(\sigma_n^E)} \pi_{q+n+1}(E_{n+1})$$

Recall that we notated $\pi_q(\Sigma^\infty X)$ as the q th stable homotopy group $\pi_q^s(X)$ (Corollary 1.1.8) of the based space X and this indeed coincides with the q th homotopy group of the suspension spectrum (Definition 1.1.5) $\Sigma^\infty X$.

1.2. Stable homotopy category.

Definition 1.2.1 ([nLa25c, Definition 0.38]). Let $f_\bullet : X_\bullet \rightarrow Y_\bullet$ be a morphism of spectra of topological spaces (Definition 1.1.4).

1. We say that f is a **strict weak equivalence** if each component $f_n : X_n \rightarrow Y_n$ is a weak homotopy equivalence (i.e. a weak equivalence for the classical model category structure for topological spaces). (♠ TODO: define the classical model category structure for topological spaces)
2. We say that f is a **strict fibration** if each component $f_n : X_n \rightarrow Y_n$ is a Serre fibration (i.e. a fibration for the classical model category structure for topological spaces). (♠ TODO: retract; relative cell complexes)
3. We say that f is a **strict cofibration** if $f_0 : X_0 \rightarrow Y_0$ and the maps

$$(f_{n+1}, \sigma_n^Y) : X_{n+1} \coprod_{S^1 \wedge X_n} (S^1 \wedge Y_n) \rightarrow Y_{n+1}$$

for $n > 1$ are retracts of relative cell complexes (i.e. cofibrations for the classical model category structure for topological spaces).

Theorem 1.2.2 ([nLa25c, Theorem 0.40]). (♠ TODO: category of spectra) The classes (i.e. strict weak equivalences, strict fibrations, and strict cofibrations) of morphisms in Definition 1.2.1 give the category of spectra the structure of a (closed) model category called the **strict/level model structure on topological spectra**, denoted by $\text{SeqSpec}(\text{Top}_{\text{cg}})_{\text{strict}}$.

Definition 1.2.3 ([nLa25c, Definition 0.14]). A morphism $f : X \rightarrow Y$ of spectra is called a **stable weak homotopy equivalence** if its image

$$\pi_\bullet(f) : \pi_\bullet(X) \xrightarrow{\sim} \pi_\bullet(Y)$$

under the stable homotopy group functor (Definition 1.1.10) is an isomorphism.

Theorem 1.2.4 ([nLa25c, Theorem 0.70]). The left Bousfield localization of the strict model structure on spectra at the class of stable weak homotopy equivalences exists.

Definition 1.2.5. The left Bousfield localization in Theorem 1.2.4 is a model category called the **stable model structure on topological spectra**, denoted by $\text{SeqSpec}(\text{Top}_{\text{cg}})_{\text{stable}}$:

$$\text{SeqSpec}(\text{Top}_{\text{cg}})_{\text{stable}} \rightleftarrows \text{SeqSpec}(\text{Top}_{\text{cg}})_{\text{strict}}.$$

The **stable homotopy category**, often denoted by $\mathcal{SH}$ or $\text{Ho}(\text{Spectra})$, is the homotopy category of the model category $\text{SeqSpec}(\text{Top}_{\text{cg}})_{\text{stable}}$.

Claim 1.2.6 ([nLa25d, Properties], [nLa25b, Proposition 4.14]). 1. The smash product of spectra makes $\mathcal{SH}$ into a symmetric monoidal category.

2. $\mathcal{SH}$ has the structure of a triangulated category, where the translation functor is the canonical suspension functor $\Sigma : \mathcal{SH} \rightarrow \mathcal{SH}$ and the distinguished triangles are the closures under isomorphisms of triangles of the images (under localization $\text{SeqSpec}(\text{Top}_{\text{cg}})_{\text{stable}} \rightarrow \mathcal{SH}$) of the canonical long homotopy cofiber sequences

$$A \xrightarrow{f} B \rightarrow \text{hocofib}(f) \rightarrow \Sigma A.$$

3. $\mathcal{SH}$ is an additive category — there is an abelian group structure on the pointed hom-sets $[X, Y]$ for X, Y in $\mathcal{SH}$.

APPENDIX A. MISCELLANEOUS DEFINITIONS

Definition A.0.1 (Topology). Let X be a set. A **topology on X** is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i \in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \dots \cap U_n$ is in $\mathcal{T}$.
Equivalently, for any $U_1, U_2 \in \mathcal{T}$, the intersection $U_1 \cap U_2$ is in $\mathcal{T}$.

If $\mathcal{T}$ is a topology on X , the pair $(X, \mathcal{T})$ is called a **topological space**. Members of $\mathcal{T}$ are called **open sets**.

A subset $C \subseteq X$ is **closed** (also called a **closed subset**) if its complement $X \setminus C$ is an open set in $\mathcal{T}$.

One very often refers to X as a topological space, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as **Top(X)**, **Top(X)**, or **Top(X)**.

Definition A.0.2. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition A.0.1). A map $f : X \rightarrow Y$ is called **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is an open set in X , that is,

$$\forall V \in \mathcal{T}_Y, f^{-1}(V) \in \mathcal{T}_X.$$

Equivalently, f is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .

A **map of topological spaces** usually refers to a continuous map between the topological spaces.

The set of continuous maps from X to Y is sometimes denoted by **C(X, Y)**. Other standard notation include **Hom_{Top}(X, Y)** or **Top(X, Y)** coming from more general notation for morphisms between objects in a category.

Definition A.0.3. Let X be a set.

1. Let τ_1, τ_2 be topologies on (Definition A.0.1) X . Say that τ_1 is **coarser than** (equivalently, **smaller than**) τ_2 if $\tau_1 \subseteq \tau_2$, and that τ_1 is **finer than** (equivalently, **larger than**) τ_2 if $\tau_2 \subseteq \tau_1$. This means that every subset of X that is open (Definition A.0.1) with respect to τ_1 is also open with respect to τ_2 .
These relations are denoted by $\tau_1 \preceq \tau_2$ for “ τ_1 coarser than τ_2 ” and $\tau_1 \succeq \tau_2$ for “ τ_1 finer than τ_2 ”; their strict versions are $\tau_1 \prec \tau_2$ and $\tau_1 \succ \tau_2$, meaning proper inclusion.
2. Let $\mathcal{C}$ be some family of topologies on X . A topology $\tau \in \mathcal{C}$ is the **coarsest** (or **smallest**) element of $\mathcal{C}$ if for every $\sigma \in \mathcal{C}$ one has $\tau \subseteq \sigma$ (equivalently, $\tau \preceq \sigma$ for all $\sigma \in \mathcal{C}$). Dually, $\tau \in \mathcal{C}$ is the **finest** (or **largest**) element of $\mathcal{C}$ if for every $\sigma \in \mathcal{C}$ one has $\sigma \subseteq \tau$ (equivalently, $\sigma \preceq \tau$ for all $\sigma \in \mathcal{C}$).

Definition A.0.4. Let X be a set and $\mathcal{S}$ a collection of subsets of X . The **topology on X generated by $\mathcal{S}$** , often denoted **$\tau(\mathcal{S})$** , is the intersection of all topologies on X that contain

$\mathcal{S}$. It is the unique coarsest topology (Definition A.0.3) on X in which every element of $\mathcal{S}$ is an open set.

The collection $\mathcal{S}$ is called a *subbasis for the topology $\tau(\mathcal{S})$* . Alternatively, one might generally call a family $\mathcal{S} \subseteq \mathcal{P}(X)$ a *subbasis on X* if $\bigcup_{S \in \mathcal{S}} S = X$ and we would consider $\mathcal{S}$ the subbasis for the topology $\tau(\mathcal{S})$.

Definition A.0.5. Let X be a set and let $\mathcal{B}$ be a collection of subsets of X . The collection $\mathcal{B}$ is called a *basis* (or *base*) for a topology (Definition A.0.1) on X if the following two conditions hold:

1. For every $x \in X$, there exists at least one $B \in \mathcal{B}$ such that $x \in B$.
2. For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Given such a collection $\mathcal{B}$, the collection $\mathcal{T}$ of all unions of elements of $\mathcal{B}$ defines a topology on X , and it coincides with $\mathcal{T}_{\mathcal{B}}$, the topology generated by $\mathcal{B}$ (Definition A.0.4). In other words,

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X : \text{for every } x \in U, \text{ there exists } B \in \mathcal{B} \text{ with } x \in B \subseteq U\}.$$

Definition A.0.6 (Pointed topological space). Let X be a topological space (Definition A.0.1) and let $x_0 \in X$ be a chosen element of X . A *pointed/based (topological) space* is a pair (X, x_0) consisting of the space X together with the distinguished point x_0 , called the *base point of X* . If the base point of a pointed space (X, x_0) is understood, then it may be suppressed from notation; in particular, X may be written as a pointed space as opposed to the full notation of (X, x_0) .

A *morphism of pointed spaces* (or *based map*) or *continuous map* between pointed spaces (X, x_0) and (Y, y_0) is a continuous map (Definition A.0.2)

$$f : X \rightarrow Y$$

such that $f(x_0) = y_0$.

The collection of pointed spaces with their morphisms form a locally small category, often called the *category of pointed spaces*. This category is often denoted by notations such as $\mathbf{Top}_*$, $\mathbf{Top}_\bullet$, $\mathbf{Top}_*$, $\mathbf{Top}_\bullet$, etc. The set of continuous maps from pointed spaces X to Y may be denoted by notations such as $C_*(X, Y)$, $C_\bullet(X, Y)$, $\mathbf{Top}_*(X, Y)$, $\mathbf{Top}_\bullet(X, Y)$, $\mathbf{Hom}_{\mathbf{Top}_\bullet}(X, Y)$, etc.

The category $\mathbf{Top}_*$ is equivalent to the full subcategory consisting of objects $(X, \{x\}, x)$ of the category $\mathbf{Top}_{\text{pair}, \bullet}$ of pointed pairs of topological spaces (Definition A.0.11).

The category $\mathbf{Top}_*$ is equivalent to the full subcategory consisting of objects of the form $(X, \{x\})$ of the category $\mathbf{Top}_{\text{pair}}$ of pairs of topological spaces.

Definition A.0.7 (Homotopy class of maps relative to a subset). Let X and Y be topological spaces (Definition A.0.1) and let $K \subseteq X$. Let $C(X, Y)$ denote the set of all continuous maps (Definition A.0.2) $f : X \rightarrow Y$.

1. Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class relative to K* or that f and g are *homotopic relative to K* if there exists a homotopy relative to K

$$H : X \times [0, 1] \rightarrow Y$$

such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and

$$H(k, t) = f(k) = g(k) \quad \text{for all } k \in K, t \in [0, 1].$$

The *homotopy class of maps relative to K* containing a map $f : X \rightarrow Y$ is denoted by $[f]_K$.

Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\emptyset$.

The *homotopy class of maps* containing a map $f : X \rightarrow Y$ is denoted by $[f]$.

The set of homotopy classes of maps may often be denoted by $[X, Y]$.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition A.0.6) and let $K \subseteq X$ be a subset containing x_0 . Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ with $f(x_0) = y_0$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class relative to K* or that f and g are *homotopic relative to K* if there exists a homotopy of based maps relative to K

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

particularly ensuring the basepoint is fixed throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

In the case that $K = \{x_0\}$, one simply says that f and g are *homotopic (as based maps)*.

The *homotopy class relative to K* containing $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]_K$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\{x_0\}$.

The *homotopy class* containing a map $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]$.

The set of homotopy classes of pointed maps $(X, x_0) \rightarrow (Y, y_0)$ may often be denoted by $[(X, x_0), (Y, y_0)]$ or by notations such as $[X, Y]_*$, $[X, Y]_\bullet$, or $[X, Y]$ if the base points are clear.

Definition A.0.8. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition A.0.1). A function $f : X \rightarrow Y$ is called a *homeomorphism* if it satisfies all of the following:

1. f is bijective;
2. f is continuous (Definition A.0.2) with respect to $\mathcal{T}_X$ and $\mathcal{T}_Y$; and
3. the inverse map $f^{-1} : Y \rightarrow X$ is also continuous.

(♠ TODO: equivalence class) If such a function exists, the spaces X and Y are said to be *homeomorphic*, or that X and Y have the *same topological type* — accordingly, the *topological type of a topological spaces* refers to its equivalence under the equivalence relation of homeomorphic spaces.

Definition A.0.9 (Compactness). Let $(X, \mathcal{T})$ be a topological space (Definition A.0.1). A subset $K \subseteq X$ is *compact* if for every collection $\{U_i\}_{i \in I}$ of open sets such that $K \subseteq \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ with $K \subseteq \bigcup_{j=1}^n U_{i_j}$.

Definition A.0.10. (♠ TODO: final topology) A topological space (Definition A.0.1) X is said to be *compactly generated* (or a *k-space*) if a subset $U \subseteq X$ is open whenever for every compact subset $K \subseteq X$, the intersection $U \cap K$ is open in the subspace K . Equivalently, X is compactly generated if and only if the topology of X is the final topology with respect to the collection of inclusions $K \hookrightarrow X$ for compact $K \subseteq X$.

Definition A.0.11. A *pointed pair of topological spaces* is a triple (X, A, x) where (X, A) is a pair of topological spaces and $x \in A$, i.e. (X, x) and (A, x) are both pointed spaces (Definition A.0.6) and the inclusion $A \hookrightarrow X$ yields a pointed map (Definition A.0.6) $(A, x) \rightarrow (X, x)$.

Definition A.0.12. A *morphism $f : (X, A, x) \rightarrow (Y, B, y)$ of pointed pairs of topological spaces* is a continuous map $f : X \rightarrow Y$ that constitutes

1. a map $(X, A) \rightarrow (Y, B)$ of pairs,
2. a map $(x, x) \rightarrow (y, y)$ of pointed spaces (Definition A.0.6), and
3. a map $(A, x) \rightarrow (B, y)$ of pointed spaces, and

In other words, f is a continuous map $X \rightarrow Y$ such that $f(A) \subseteq B$ and $f(x) = y$.

Definition A.0.13. The *category of pairs of topological spaces*, often denoted by notations such as **Top**_{*,pair}, **Top**_{•,pair}, **Top**_{*,rel}, or **Top**_{•,rel} (rel for relative), is the category defined by:

- **Objects:** Pointed Pairs (Definition A.0.11) (X, A, x) of topological spaces.
- **Morphisms:** Maps $f : (X, A, x) \rightarrow (Y, B, y)$ of pointed pairs (Definition A.0.12).
- **Composition:** The composition of morphisms is the usual composition of continuous maps.
- **Identity:** For each object (X, A, x) , the identity morphism is the identity map $\text{id}_X : (X, A, x) \rightarrow (X, A, x)$.

It is a locally small category.

Definition A.0.14. An *n-tuple of topological spaces* is a sequence $(X_1, X_2, \dots, X_n)$ of topological spaces (Definition A.0.1) such that $X_1 \supseteq X_2 \supseteq \dots \supseteq X_n$.

Definition A.0.15. A *morphism $f : (X_1, \dots, X_n) \rightarrow (Y_1, \dots, Y_n)$ of n-tuples of topological spaces* is a continuous map $f : X_1 \rightarrow Y_1$ that constitutes a map $(X_i, X_{i+1}) \rightarrow (Y_i, Y_{i+1})$ of pairs for each $i \in \{1, \dots, n-1\}$. In other words, f is a continuous map $X_1 \rightarrow Y_1$ such that $f(X_i) \subseteq Y_i$ for all $i = 1, \dots, n$.

Definition A.0.16. The *category of n -tuples of topological spaces*, often denoted by notations such as $\mathbf{Top}^n$ (not to be confused with the n -time self product of the category $\mathbf{Top}$ of topological spaces), is the category defined by:

- **Objects:** n -tuples (Definition A.0.14) $(X_1, \dots, X_n)$ of topological spaces.
- **Morphisms:** Morphisms of n -tuples (Definition A.0.15).
- **Composition:** The composition of morphisms is the usual composition of continuous maps.
- **Identity:** For each object $(X_1, \dots, X_n)$, the identity morphism is the identity map id_{X_1} .

It is a locally small category.

Definition A.0.17. A *pointed n -tuple of topological spaces* is a $(n+1)$ -tuple $(X_1, X_2, \dots, X_n, x)$ where $(X_1, X_2, \dots, X_n)$ is an n -tuple of topological spaces (Definition A.0.14) and $x \in X_n$, i.e., each (X_i, x) is a pointed space (Definition A.0.6) and the inclusions $X_{i+1} \hookrightarrow X_i$ yield pointed maps (Definition A.0.6) $(X_{i+1}, x) \rightarrow (X_i, x)$ for all $1 \leq i < n$.

Definition A.0.18. A *morphism $f : (X_1, \dots, X_n, x) \rightarrow (Y_1, \dots, Y_n, y)$ of pointed n -tuples of topological spaces* is a continuous map $f : X_1 \rightarrow Y_1$ that constitutes

1. a morphism of n -tuples (Definition A.0.15) $(X_1, \dots, X_n) \rightarrow (Y_1, \dots, Y_n)$, and
2. a pointed map (Definition A.0.6) $(X_i, x) \rightarrow (Y_i, y)$ for each $i \in \{1, \dots, n\}$.

In other words, f is a continuous map $X_1 \rightarrow Y_1$ such that $f(X_i) \subseteq Y_i$ for all i and $f(x) = y$.

Definition A.0.19. The *category of pointed n -tuples of topological spaces*, often denoted by notations such as $\mathbf{Top}_*^n$ or $\mathbf{Top}_*^n$ (not to be confused with the n -time self product of the category $\mathbf{Top}_*$ (Definition A.0.6) of topological spaces), is the category defined by:

- **Objects:** Pointed n -tuples (Definition A.0.17) $(X_1, \dots, X_n, x)$ of topological spaces.
- **Morphisms:** Morphisms of pointed n -tuples (Definition A.0.18).
- **Composition:** The composition of morphisms is the usual composition of continuous maps.
- **Identity:** For each object $(X_1, \dots, X_n, x)$, the identity morphism is the identity map id_{X_1} .

It is a locally small category.

Definition A.0.20 (Homotopy groups of a pointed topological space). For any pointed topological space (Definition A.0.6) (X, x_0) and integer $n \geq 0$, the *n -th homotopy set of X at x_0* , denoted $\pi_n(X, x_0)$ or simply $\pi_n(X)$ when the base point x_0 is understood or a choice of base point does not really matter, is defined as the set of all homotopy classes (rel. ∂I^n) (Definition A.0.7) of paired maps

$$f : (I^n, \partial I^n) \rightarrow (X, \{x_0\}),$$

where $I^n = [0, 1]^n$ and ∂I^0 is taken to be the singleton I^0 instead of the literal boundary thereof. Equivalently, one can also define it as the set of all homotopy classes of pointed

maps (Definition A.0.6)

$$f : (S^n, s_0) \rightarrow (X, x_0),$$

where S^n is the n -sphere and $s_0 \in S^n$ is a choice of base point. In other words, $\pi_n(X, x_0)$ is by definition $[(S^n, s_0), (X, x_0)]_*$ (Definition A.0.7).

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of pointed maps, and for $n \geq 2$, it is abelian. As such, for $n \geq 1$, it is appropriate to refer to $\pi_n(X, x_0)$ as the *n th homotopy group of (X, x_0)* .

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

For every $n \geq 0$, the assignment $(X, x_0) \mapsto \pi_n(X, x_0)$ is functorial — in fact

1. For $n = 0$, it is a functor $\text{Top}_* \rightarrow \mathbf{Sets}$,
2. For $n = 1$, it is a functor $\text{Top}_* \rightarrow \mathbf{Grp}$,
3. For $n \geq 2$, it is a functor $\text{Top}_* \rightarrow \mathbf{Ab}()$.

Given a map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces, the induced map $\pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$ is often denoted by f_* .

The homotopy set $\pi_n(X, x_0)$ is naturally isomorphic to the relative homotopy set $\pi_n(X, \{x_0\}, x_0)$ (Definition A.0.21).

We further note that the set $\pi_n(X, x_0) = [(S^n, s_0), (X, x_0)]_*$ is identifiable with $\pi_0(\text{Map}_*((S^n, s_0), (X, x_0)))$ (Proposition 1.0.10).

Definition A.0.21 (Relative homotopy groups of pointed pairs). For any pointed pair of topological spaces (Definition A.0.11) (X, A, x_0) and an integer $n \geq 1$, the *n -th relative homotopy set of (X, A) at x_0* , denoted $\pi_n(X, A, x_0)$, is defined as the set of all homotopy classes (rel. J^{n-1}) (Definition A.0.7) of maps of triples (Definition A.0.14)

$$f : (I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, \{x_0\}),$$

where $I^n = [0, 1]^n$ is the n -cube, ∂I^n is its boundary, and

$$J^{n-1} := \overline{\partial I^n \setminus (I^{n-1} \times \{1\})} = (I^{n-1} \times \{0\}) \cup (\partial I^{n-1} \times I) \subseteq \partial I^n.$$

Explicitly, an element of $\pi_n(X, A, x_0)$ is represented by a continuous map (Definition A.0.2) $f : I^n \rightarrow X$ such that:

1. $f(\partial I^n) \subseteq A$,
2. $f(J^{n-1}) = \{x_0\}$.

Equivalently, $\pi_n(X, A, x_0)$ can be defined as the set of all homotopy classes of maps of pairs

$$f : (D^n, S^{n-1}, s_0) \rightarrow (X, A, x_0),$$

where D^n is the n -disk, S^{n-1} is its boundary sphere, and $s_0 \in S^{n-1}$ is a base point.

For $n \geq 2$, $\pi_n(X, A, x_0)$ is a group under the concatenation of maps in the first coordinate, and for $n \geq 3$, it is abelian. As such:

1. For $n = 1$, the assignment $(X, A, x_0) \mapsto \pi_1(X, A, x_0)$ is a functor $\mathbf{Top}_{*,\text{pair}} \rightarrow \mathbf{Sets}$,
2. For $n = 2$, it is a functor $\mathbf{Top}_{*,\text{pair}} \rightarrow \mathbf{Grp}$,
3. For $n \geq 3$, it is a functor $\mathbf{Top}_{*,\text{pair}} \rightarrow \mathbf{Ab}$.

For $n \geq 2$, $\pi_n(X, A, x_0)$ is called the *n -th relative homotopy group of (X, A) at x_0* . Given a map $f : (X, A, x_0) \rightarrow (Y, B, y_0)$ of pointed pairs, the induced map $\pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$ is denoted by f_* .

Definition A.0.22 (Disjoint union of sets). Let $\{X_i\}_{i \in I}$ be an arbitrary small collection of sets. The *disjoint union* of the collection, denoted by $\coprod_{i \in I} X_i$ or $\sqcup_{i \in I} X_i$, is defined as the set

$$\coprod_{i \in I} X_i := \bigcup_{i \in I} \{(x, i) : x \in X_i\}.$$

The indices $i \in I$ serve as "tags" to ensure that the contributing sets are treated as disjoint, even if their underlying elements are the same. For each $j \in I$, there is a canonical injection

$$\iota_j : X_j \hookrightarrow \coprod_{i \in I} X_i, \quad x \mapsto (x, j).$$

The disjoint union is the coproduct in the category of sets. For a finite collection $X_1, \dots, X_n$, it is often denoted $X_1 \amalg \dots \amalg X_n$.

Definition A.0.23 (Disjoint union topology). Let $\{X_i\}_{i \in I}$ be an arbitrary small collection of topological spaces (Definition A.0.1). The *disjoint union space* (or *topological sum*), denoted by $\coprod_{i \in I} X_i$, is the set-theoretic disjoint union (Definition A.0.22) $\bigcup_{i \in I} (X_i \times \{i\})$ endowed with the *disjoint union topology*, which is the finest topology (Definition A.0.3) such that each canonical injection

$$\iota_j : X_j \hookrightarrow \coprod_{i \in I} X_i, \quad x \mapsto (x, j)$$

is continuous (Definition A.0.2).

Specifically, a subset $U \subseteq \coprod_{i \in I} X_i$ is open (Definition A.0.1) if and only if $\iota_i^{-1}(U)$ is open in X_i for every $i \in I$. The disjoint union space is the coproduct in the category of topological spaces.

Definition A.0.24 (Quotient topology). Let X be a topological space (Definition A.0.1), Y be a set, and $q : X \rightarrow Y$ be a surjective function. The *quotient topology on Y induced by q* is the finest topology (Definition A.0.3) on Y such that q is continuous (Definition A.0.2).

Equivalently, a subset $V \subseteq Y$ is open in the quotient topology if and only if its preimage $q^{-1}(V)$ is open in X . The pair (Y, q) is then called a *quotient space* of X .

Equivalently, if $\sim$ is an equivalence relation on X , the quotient space $X/\sim$ is the set of equivalence classes endowed with the quotient topology induced by the canonical projection $\pi : X \rightarrow X/\sim$.

Definition A.0.25 (Wedge sum). Let $\{(X_i, x_i)\}_{i \in I}$ be a collection of pointed topological spaces (Definition A.0.6). The *wedge sum of the collection*, denoted by $\bigvee_{i \in I} X_i$, is the quotient (Definition A.0.24) of the disjoint union (Definition A.0.23) $\coprod_{i \in I} X_i$ by the equivalence relation that identifies all base points $\{x_i\}_{i \in I}$ to a single point.

For a finite collection $(X_1, x_1), \dots, (X_n, x_n)$, the wedge sum is denoted $X_1 \vee \dots \vee X_n$. It is the coproduct in the category of pointed topological spaces (Definition A.0.6).

Definition A.0.26 (Smash product). Let (X, x_0) and (Y, y_0) be pointed topological spaces (Definition A.0.6). The *smash product of X and Y* , denoted by $X \wedge Y$, is the quotient space (Definition A.0.24)

$$X \wedge Y := (X \times Y) / (X \vee Y),$$

where $X \vee Y$ is viewed as the subspace $(X \times \{y_0\}) \cup (\{x_0\} \times Y)$ of the product space $X \times Y$.

The base point of $X \wedge Y$ is the equivalence class of $X \vee Y$. The smash product is the unique pointed space such that maps $X \wedge Y \rightarrow Z$ correspond naturally to pointed maps (Definition A.0.6) $X \times Y \rightarrow Z$ that are constant on the wedge sum $X \vee Y$.

Definition A.0.27. Let X and Y be topological spaces (Definition A.0.1).

1. Write $\text{Open}(Y)$ for the topology (Definition A.0.1) of Y and $\text{Comp}(X)$ for the family of compact subsets of X . For $K \in \text{Comp}(X)$ and $U \in \text{Open}(Y)$, define the subset

$$\langle K, U \rangle := \{ f \in C(X, Y) \mid f(K) \subseteq U \} \subseteq C(X, Y).$$

where $C(X, Y)$ (Definition A.0.2) is the set of all continuous maps (Definition A.0.2) $X \rightarrow Y$. The collection of all such subsets is denoted by $\mathcal{S}_{\text{co}}(X, Y)$.

2. The *compact-open topology on $C(X, Y)$* (Definition A.0.2) is the topology (Definition A.0.1) generated by (Definition A.0.4) the subbasis (Definition A.0.4) $\mathcal{S}_{\text{co}}(X, Y)$. The resulting topological space is called the *function space from X to Y* and is commonly denoted by notations such as $\text{Map}(X, Y)$, Y^X , $\text{Fun}(X, Y)$, or $\mathbf{F}(X, Y)$. As a set, it equals $C(X, Y)$.
3. Let (X, x_0) and (Y, y_0) be based topological spaces (Definition A.0.6). The *based function space from (X, x_0) to (Y, y_0)* is the subspace of $\text{Map}(X, Y)$ consisting of all based maps f with $f(x_0) = y_0$, equivalently $C_*((X, x_0), (Y, y_0))$ endowed with the subspace topology inherited from the compact-open function space $\text{Map}(X, Y)$. It itself is a based topological space whose base point is given by the constant map $X \rightarrow Y$ with value y_0 . Common notations for the based function space include those which include a star or bullet such as $\text{Map}_*((X, x_0), (Y, y_0))$, $\text{Map}_*(X, Y)$, or Y_*^X to emphasize that the spaces involved are pointed, or those which omit a star or bullet, such as $\text{Map}(X, Y)$, and Y^X .

Definition A.0.28 (Reduced suspension). Let (X, x_0) be a pointed topological space (Definition A.0.6). The *reduced suspension of X* , denoted by ΣX , is the smash product (Definition A.0.26) of X with the circle S^1 : (♠ TODO: circle)

$$\Sigma X := X \wedge S^1.$$

Equivalently, ΣX is the quotient of the unreduced suspension SX by the image of the segment $\{x_0\} \times [0, 1]$. As a quotient of $X \times [0, 1]$, the relation identifies all points in the set $(X \times \{0\}) \cup (X \times \{1\}) \cup (\{x_0\} \times [0, 1])$ to a single base point.

In particular, Σ is a functor $\mathbf{Top}_* \rightarrow \mathbf{Top}_*$.

The n -fold reduced suspension of (X, x_0) is denoted by $\Sigma^n X$.

Definition A.0.29 (Free path space of a topological space). Let Y be a topological space (Definition A.0.1). The *free path space of Y* , is the function space (Definition A.0.27) $\text{Map}(I, Y)$ of all paths $\gamma : [0, 1] \rightarrow Y$ equipped with the compact-open topology (Definition A.0.27). As such, it is also appropriate to denote the free path space of Y as Y^I .

Definition A.0.30 (Free loop space of a topological space). Let Y be a topological space (Definition A.0.1).

The *free loop space*, denoted $\mathcal{LY}$, is the mapping space (Definition A.0.27) $\text{Map}(S^1, Y)$ of maps from the circle to Y with no basepoint restriction.

Definition A.0.31 (Based loop space). Let (Y, y_0) be a pointed topological space (Definition A.0.6). The *based loop space of Y* , denoted by ΩY , is the based mapping space (Definition A.0.27) $\text{Map}_*(S^1, Y)$.

Explicitly, ΩY is the subspace of $\text{Map}(I, Y)$ consisting of paths $\gamma : [0, 1] \rightarrow Y$ such that $\gamma(0) = \gamma(1) = y_0$. The base point of ΩY is the constant loop at y_0 .

In particular, Ω is a functor $\mathbf{Top}_* \rightarrow \mathbf{Top}_*$.

The n -fold loop space of X is denoted by $\Omega^n X$.

Proposition A.0.32. For all pointed topological spaces (Definition A.0.6) X, Y , there is an adjunction between the functors Σ (Definition A.0.28) and Ω (Definition A.0.31). More specifically, there are homeomorphisms (Definition A.0.8)

$$\text{Map}_*(\Sigma X, Y) \cong \text{Map}_*(X, \Omega Y)$$

of pointed topological spaces (Definition A.0.27) natural in X and Y .

Applying π_0 , we have bijections

$$[\Sigma X, Y]_* \cong [X, \Omega Y]_*.$$

(Definition A.0.7) In fact, these homotopy class sets have group structures (Definition A.0.20) and the bijection is actually a group isomorphism.

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